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Materials Design for Compute Thermal Management From Dielectric Fluids to Gallium-Based Cooling Systems

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Materials Design for Compute Thermal Management: From Dielectric Fluids to Gallium-Based Cooling Systems

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Abstract

Nearly all electricity consumed by compute infrastructure exits the facility as thermal energy. Bitcoin mining consumes 138–211 TWh annually; AI data centers are projected to exceed 1,000 TWh by end of 2026. This article examines compute thermal management as a materials design problem. Current first-generation immersion cooling uses dielectric fluids (thermal conductivity 0.06–0.14 W/m·K) that capture waste heat at 65–75°C, directly usable for district heating without heat pumps. Six operational deployments are documented, with lifecycle CO_2 displacement estimated at 2,000–3,800 tonnes per 1.7 MW container per year at 70–85% thermal utilization. Second-generation gallium-based liquid metal systems achieve thermal conductivities of 26–39 W/m·K at the thermal interface, a 200–600× improvement, with experimentally demonstrated 60–68% temperature reduction under pulsed thermal loads. Third-generation embedded two-phase microfluidic cooling, demonstrated under DARPA's ICECool program at 1 kW/cm², points toward chip-level thermal management at heat flux densities that facility-level immersion cannot reach. Each generation of cooling material determines the temperature, quality, and reusability of the waste heat produced. The materials choice is therefore not merely a cooling decision but an energy-systems design decision with consequences for water consumption and carbon displacement. This article reviews the materials science of each generation, documents the engineering challenges (corrosion, electrical conductivity, surface tension, supply chain concentration), and argues that the trajectory from dielectric fluids to gallium-based systems to embedded phase-change cooling represents a coherent materials design frontier with facility-scale thermodynamic implications.

Keywords – Thermal management; Materials design; Immersion cooling; Gallium; Liquid metals; Phase-change materials; Dielectric fluids; Waste heat recovery; Data centers; District heating

1. Introduction: The Thermal Problem as a Materials Problem

In January 2026, a shipping container arrived in a small town in Scandinavia. Inside were 160 Bitcoin mining rigs submerged in dielectric fluid, generating 1.5 MW of thermal energy that was pumped into the town's district heating grid, warming approximately 2,000 homes (assuming ~5 MWh/home/year heating demand). The company that built it, Latvia-based Power Mining, described itself as a heating company that happens to mine Bitcoin [5]. Simultaneously, a greenhouse in Manitoba was being heated by 360 liquid-cooled mining servers capturing approximately 90% of consumed electricity as heat at 75°C [6]. And in Dublin, students at the Technological University attended lectures in buildings heated by waste heat from an Amazon Web Services data center [15].

These deployments share a common engineering feature: the cooling material determines the output temperature, and the output temperature determines the heat's industrial utility. Air-cooled systems reject heat at 25–35°C—too low for district heating networks, which require 60–80°C supply temperatures [4]. Immersion in dielectric fluid raises the capture temperature to 65–75°C, crossing the district heating threshold. Gallium-based liquid metals, with thermal conductivities two to three orders of magnitude higher than dielectric fluids, could raise output temperatures above 80°C while simultaneously managing the escalating heat flux densities of AI accelerators. The choice of cooling material is therefore an energy-systems design decision, not merely a thermal engineering one.

The data center waste heat recovery literature has grown substantially in recent years. Huang et al. [33] reviewed data centers as energy prosumers in district heating systems; Wahlroos et al. [34] analyzed waste heat utilization in Nordic data centers; Yuan et al. [35, 36] published comprehensive reviews of data center waste heat recovery technologies and their integration with district heating networks. These studies approach the problem from the system and network perspective. The present article complements this literature by approaching the problem from the cooling *material* perspective: examining how the thermal properties of the cooling medium itself determine the temperature, quality, and reusability of waste heat, and tracing a three-generation materials progression that the existing literature has not framed.

This article examines three generations of compute cooling materials—dielectric fluids, gallium-based liquid metals, and embedded two-phase microfluidic systems—through the lens of materials design. It documents operational deployments that validate first-generation technology, reviews the materials science frontier of second- and third-generation systems, and identifies the engineering challenges that separate laboratory demonstrations from facility-scale deployment.

The energy baseline is drawn from the Cambridge Centre for Alternative Finance's Digital Mining Industry Report (April 2025), which surveyed 49 firms covering 48% of global hash rate [1]: Bitcoin mining consumes 138 TWh/yr (upper estimate 211 TWh [2]) with a sustainable energy share of 52.4% (42.6% renewables, 9.8% nuclear). AI data centers are projected by the IEA to exceed 1,000 TWh by end of 2026, more than 7× Bitcoin mining's consumption [3].

Selection bias note: The Cambridge survey's respondents are disproportionately publicly listed (41%) and North American. The unreported 52% includes operations in Central Asia and Africa where coal and gas typically dominate. The true global sustainable share may be materially lower than 52.4% [1].

2. Thermodynamic Framework

Every processor converts electricity into computation and heat at a ratio of approximately 100%. The first law of thermodynamics guarantees energy conservation; the second law constrains its quality. Electricity entering a facility is high-grade energy; heat exiting at 65°C is low-grade, suitable for space and water heating but not for mechanical or electronic work. The energy is conserved; the exergy is reduced.

A heat pump delivers 3–4 joules of thermal energy per joule of electricity consumed (COP of 3–4). A mining rig delivers approximately 1 joule of heat per joule of electricity, plus cryptocurrency revenue. Mining-as-heating is therefore thermodynamically inferior to a heat pump for pure heating purposes and is economically competitive only when computation revenue offsets the COP gap. The materials question is how to maximize the temperature and quality of the rejected heat so that the gap narrows and the co-product value increases.

3. First-Generation Cooling Materials: Dielectric Fluids

3.1. Thermal properties and performance envelope

Single-phase immersion fluids—engineered hydrocarbons or synthetic esters—offer thermal conductivities of 0.06–0.14 W/m·K, specific heat capacities of 1.5–2.1 kJ/kg·K, and operating ranges from –40°C to +200°C [9, 10, 11]. They are chemically inert, electrically non-conductive, and non-flammable.

Two-phase fluorocarbon fluids offer higher heat transfer coefficients through phase-change latent heat, but several formulations fall under PFAS classification; the EU's proposed PFAS restriction under REACH may limit future availability [12, 13]. Non-PFAS alternatives (hydrocarbon-based, ester-based, nanoparticle-enhanced) are under development but have not achieved performance parity in published benchmarks. The broader liquid cooling market for AI data centers is projected to grow from $2.8 billion in 2025 to over $21 billion by 2032 at 30%+ CAGR, underscoring the industrial scale of this materials challenge.

The EU Energy Efficiency Directive (2023/1791) now mandates waste heat recovery for new data centers above 1 MW within the European Union [13]. This regulatory shift changes the question from whether data centers should recover heat to *how* they should recover it—and the "how" is precisely where the materials framework developed in this article becomes policy-relevant.

Single-phase fluids exhibit measurable total acid number (TAN) increases after 8,000–12,000 hours of continuous operation above 60°C, indicating oxidative degradation. Replacement intervals of 5–8 years are typical for closed-loop systems with adequate oxygen exclusion [9, 10, 11]. Operational complexity at scale—leak detection, fluid quality monitoring, modified fire suppression, and maintenance retraining—represents substantial transition cost for facilities migrating from air-cooled architectures.

3.2. The temperature gradient breakthrough

District heating networks require supply temperatures of 60–80°C [4]. Air-cooled data centers produce waste heat at 25–35°C, below this threshold. Bridging the gap requires heat pumps ($500K–$2M additional capital) that consume additional electricity. This temperature gap has historically limited data center heat reuse to a small number of purpose-built facilities.

Immersion cooling changes this fundamentally. Submerging hardware in dielectric fluid captures heat at 65–75°C—directly within the district heating range. No heat pump is required. Power Mining's containers produce heat at 65°C; Canaan's system produces water at 75°C [5, 6]. The same material property that solves the cooling problem simultaneously solves the heat-reuse problem. This is the article's central materials-design insight: cooling material selection determines not only thermal performance but downstream energy-system integration.

3.3. Water elimination

Evaporative cooling, the dominant paradigm for large, air-cooled data centers, consumes approximately 110 million gallons of freshwater annually for a medium facility (~10 MW), with 80% permanently lost through evaporation [7]. For an equivalent 10 MW immersion-cooled facility, water consumption is zero—dielectric fluid circulates in a sealed closed loop with no evaporative loss. Each 100-word AI prompt processed by an evaporatively cooled data center consumes approximately 519 mL of water [8]. At scale, U.S. data centers collectively consume an estimated 163.7 billion gallons annually [7]. Immersion cooling represents a complete elimination of freshwater demand from the cooling system. This claim refers to the cooling loop itself; the water footprint of electricity generation (hydropower, thermal plant cooling) that powers the compute is a separate lifecycle consideration not addressed here.

4. Engineering Validation: Operational Deployments

4.1. Documented heat-reuse systems

Six operational deployments validate the first-generation materials approach. The Dublin AWS scheme is the only deployment with independently measured emissions data: 704 tCO_2 abated in 2024, verified by TU Dublin [15]. For Bitcoin mining deployments, environmental claims are based on engineering projections from reported specifications, not measured outcomes. These six cases are illustrative examples within a broader trend; Yuan et al. [35] documented over 20 operational or planned data center heat-reuse installations across Europe. Key deployments include: Power Mining (Scandinavia)—160 immersed rigs, 1.7 MW thermal, 65°C output, approximately 2,000 homes heated; Canaan/Bitforest (Manitoba)—360 liquid-cooled servers, 75°C output, greenhouse heating capturing approximately 90% of consumed electricity [5, 6]; MintGreen (Lonsdale, BC)—providing 96% of a commercial building's heating demand from mining waste heat [14]; and AWS Dublin—AI/cloud waste heat piped to a university campus with third-party-verified CO_2 measurement [15].

4.2. Lifecycle CO_2 displacement model

The following model estimates annual CO_2 displacement for a single Power Mining container (1.7 MW thermal output, 65°C, operating in a Nordic climate with 8-month heating season). This is an engineering projection based on published specifications, not a measured result. Thermal output of 1.7 MW continuous over approximately 5,840 annual operating hours yields 9,928 MWh of delivered thermal energy. In practice, thermal utilization rates of 70–85% are typical for well-integrated district heating connections due to demand fluctuation, maintenance downtime, and network constraints, reducing effective delivery to approximately 6,950–8,440 MWh. Depending on the displaced fuel—natural gas (0.2 tCO_2/MWh), fuel oil (0.27 tCO_2/MWh), or peat (0.38 tCO_2/MWh)—annual displacement ranges from approximately 1,400 to 3,200 tonnes of CO_2 per container after accounting for utilization losses. The electricity consumed by the mining rigs carries its own emissions burden; at Bitcoin's current energy mix (52.4% sustainable), each MWh of mining electricity produces approximately 0.29 tCO_2. The net climate benefit is the difference between displaced heating emissions and mining electricity emissions.

4.3. Scalability and geographic constraints

Finland, Sweden, Denmark, and Norway deliver more than 50% of heating through district networks. Compute heat integration works in these regions because of cold climate (8–10 months demand), existing pipe infrastructure, supportive municipal policy, and high conventional energy costs [4]. Replication outside Scandinavia faces significant barriers: most North American and Asian cities lack district heating infrastructure, with construction costs of $5–20M per kilometer. Year-round applications—greenhouses, aquaculture, industrial process heat—extend viable geography without eliminating the seasonal constraint. A further limitation: many Finnish district heating systems already source heat from co-generation plants. In these systems, the marginal climate benefit of mining heat is reduced because the displaced source was already efficient [37]. The strongest case exists where networks depend on standalone fossil boilers.

District heating networks are designed for 30-year lifespans. Mining hardware is obsolete in 3–5 years; data center hardware in 7–10. Municipalities evaluating mining-heated districts must plan for multiple hardware refresh cycles per infrastructure lifecycle [18]. This does not invalidate the model but introduces planning complexity that conventional heat sources do not.

It is worth noting that Bitcoin mining also provides rapid demand response that no other large-scale compute workload can match. Riot Platforms' 700 MW Texas facility curtailed more than 95% of power usage during the August 2023 heat wave, earning $31.7M in ERCOT curtailment credits in a single month [S1, S2]. ERCOT large-load requests reached 226 GW in 2025, with AI companies now comprising 73% of new applications. The flexible-load model that miners built is being inherited by an industry seven times larger. The demand response dimension is treated in detail in the Supplementary Material (Section S1).

Table 1. Three-Generation Materials Comparison

Property	1st Gen: Dielectric Fluids	2nd Gen: Ga-Based Liquid Metals	3rd Gen: Embedded Two-Phase
Thermal conductivity	0.06–0.14 W/m·K	26–39 W/m·K (TIM); loop TBD	Fluid-dependent; heat removal >1 kW/cm²
Output temperature	65–75°C	>80°C (projected)	60–80°C (tunable via pressure)
Electrical conductivity	Non-conductive (safe)	~3.5×10^6 S/m (hazard)	Non-conductive (dielectric refrigerant)
Deployment scale	Facility-level (MW)	TIM: consumer (PS5); Loop: lab only	Chip-level (demonstrated); facility TBD
TRL	8–9 (commercial)	TIM: 8–9; Loop: 3–4	4–5 (DARPA demo)
Water consumption	Zero (closed loop)	Zero (closed loop)	Zero (closed loop)
Key challenge	Low thermal conductivity; fluid degradation	Al corrosion; electrical containment; supply chain	Manufacturing scalability; microchannel reliability
DH compatibility	Direct (no heat pump)	Direct + higher T (projected)	Direct (pressure-tunable)

Table 1. Three-generation materials comparison for compute cooling. TRL = Technology Readiness Level. DH = District Heating. TIM applications (milligrams, chip-level) and cooling loop applications (kilograms, facility-level) are distinguished for Ga-based systems, reflecting different readiness levels.

5. Second-Generation Materials: Gallium-Based Liquid Metal Systems

5.1. Thermal performance

The dielectric fluids documented in Section 3 represent the first generation of compute liquid cooling. A second generation is emerging from materials science: gallium-based liquid metal alloys. Pure gallium exhibits a thermal conductivity of 33.7 W/m·K in the solid state and 24 W/m·K as a liquid, with a latent heat of 80,300 J/kg at a melting point of 29.8°C [19]. This represents a 200–600× improvement in thermal conductivity over current dielectric fluids. GaInSn eutectic alloys (Galinstan: 68.5% Ga, 21.5% In, 10% Sn) achieve thermal conductivities of 26–39 W/m·K as thermal interface materials, with optimized formulations reaching higher values depending on composition, bond line thickness, surface roughness, and contact pressure [20, 21].

It is essential to distinguish two applications of gallium-based materials with very different engineering requirements and readiness levels. Ga-based *thermal interface materials* (TIMs), applied at the die-to-heat-sink junction in milligram quantities, are commercially mature: Sony's PlayStation 5 uses a Galinstan-based TIM in mass production, and Indium Corporation's GalliTHERM product line targets HPC semiconductor applications (TRL 8–9). Ga-based *cooling loops*, circulating kilograms of liquid metal through meters of facility piping, are essentially undemonstrated at scale (TRL 3–4). Bridging this gap—from milligrams at the chip to kilograms in facility piping—is the central scaling challenge for the second generation.

5.2. Phase-change properties and figures of merit

The performance advantage of gallium extends beyond thermal conductivity. Boteler et al. demonstrated that for a ΔT of 20°C, gallium absorbs 7× more energy per unit volume than copper and 3× more than organic paraffin phase-change materials, owing to its combination of high thermal conductivity and substantial latent heat [19]. Shao et al. developed a figure-of-merit (FOM) for PCM cooling capacity defined as the product $\kappa_L \times \rho \times h$ (thermal conductivity × density × latent heat); metallic alloys achieve FOMs 1–2 orders of magnitude higher than organic PCMs ($2{,}603 \times 10^3$ $W^2s/(Km^4)$ for Bi-based metallic alloy vs. 22×10^3 for paraffin wax) [22].

5.3. Experimental validation under pulsed thermal loads

Gonzalez-Nino et al. experimentally demonstrated that metallic PCMs reduced temperature rise by 60–68% under 19 ms pulsed thermal loads at heat fluxes up to 338 W/cm², compared to only 14–17% reduction for the best-performing organic PCM [23]. At 160 W pulse power, the metallic PCM maintained a temperature within 5°C of its melting point while the dielectric baseline rose 119°C—the metallic PCM absorbed pulses 3–4× more powerful at equivalent temperature. These results are directly relevant to the pulsed and bursty thermal profiles characteristic of AI training workloads, where heat flux transients can exceed steady-state averages by an order of magnitude.

Ma and Liu proposed the concept of nano liquid-metal fluid—gallium-based coolants enhanced with conductive nanoparticles—demonstrating thermal conductivity enhancement ratios up to 2.3× with 20% carbon nanotube volume fraction in liquid gallium [24]. This result, while promising, has not been reproduced at scale; CNT dispersion in liquid gallium faces sedimentation, agglomeration, and oxide entrapment challenges at volumes beyond the laboratory. The liquid metal TIM market reached $305.9 million in 2025, driven primarily by AI accelerator thermal demands, and is projected to grow at 6.6% CAGR to $479.7 million by 2032 [25].

6. Engineering Challenges for Gallium-Based Cooling Systems

6.1. Corrosion and material compatibility

Gallium alloys aggressively attack aluminum via liquid metal embrittlement, requiring complete exclusion of aluminum from any cooling loop. Deng and Liu tested liquid gallium against four metal substrates at temperatures relevant to chip cooling and found that 6063 aluminum alloy suffered complete gallium infiltration within five hours at 120°C, with SEM/EDS analysis revealing gallium concentrations of 25–35% throughout the corrosion diffusion zone [29]. T2 copper alloy exhibited slow, general corrosion at a rate of approximately 30–60 micrometers per month at 60°C, with corrosion products that could shed and block flow channels in a dynamic system. By contrast, 1Cr18Ni9 stainless steel showed zero detectable gallium infiltration after 30 days of continuous exposure at 60°C, confirming its suitability as the structural material of choice despite its lower thermal conductivity [29]. For compute cooling applications, the practical materials palette is nickel-plated copper or 316L stainless steel throughout the cooling loop [24].

6.2. Electrical conductivity and containment

Unlike dielectric fluids, liquid metals are electrically conductive (approximately 3.5×10^6 S/m for gallium), demanding rigorous containment and leak-detection infrastructure. A dielectric fluid leak is a maintenance event; a liquid metal leak near active electronics is a catastrophic short-circuit risk. This constraint fundamentally changes facility design requirements relative to current immersion cooling, where hardware is submerged directly in the cooling medium. Gallium-based systems require sealed, isolated cooling channels rather than open-bath immersion.

6.3. Surface tension and wettability

Gallium exhibits high surface tension (approximately 709 mN/m vs. approximately 73 mN/m for water [24]), which limits wettability on untreated surfaces and requires oxide-layer engineering for uniform thermal contact. In practice, a native gallium oxide film forms rapidly on exposed surfaces, and controlled management of this oxide layer—through surface treatments, reduced-oxygen environments, or chemical reduction—is necessary to achieve consistent thermal interface performance.

6.4. Supercooling mitigation

A separate engineering challenge for gallium as a phase-change thermal storage medium is its pronounced supercooling: bulk gallium can remain liquid to approximately −38°C, nearly 68°C below its 29.8°C melting point. Zhang et al. demonstrated that adding 0.5 wt.% TeO_2 particles, whose lattice constants closely match those of α-Ga, reduced gallium's supercooling by 44%, from 67.8°C to 38.2°C [30]. Mitigating supercooling is essential if gallium-based PCMs are to function reliably in seasonal thermal storage applications where predictable freeze–thaw cycling at the design temperature is required.

6.5. Toxicity considerations

Gallium itself exhibits low toxicity and good biocompatibility. However, Galinstan alloys contain approximately 21.5% indium, and indium compounds have documented pulmonary toxicity (indium lung disease) associated with occupational exposure during semiconductor manufacturing. At facility scale, handling protocols for indium-containing alloys must address inhalation risks from aerosolized particles during maintenance, leak cleanup, and alloy preparation. This does not preclude deployment but requires occupational health planning beyond what dielectric fluids demand.

6.6. Implications for heat-reuse systems

For the heat-reuse framework developed in this article, gallium-based cooling has a direct implication: a gallium cooling loop could capture waste heat at higher temperatures with lower thermal resistance than hydrocarbon-based systems, potentially enabling output temperatures above 80°C. This is a projection based on the reduced thermal resistance at the chip-coolant interface, not yet demonstrated in district heating applications. Higher output temperatures would expand the range of industrial heat applications—including absorption chilling, food processing, and low-temperature industrial drying—accessible from compute waste heat. If Ga-based systems can deliver output temperatures consistently above 80°C, they may eliminate the need for heat pump boosting that some immersion-cooled installations still require for network compatibility.

A complementary design principle is time-scale matched phase-change materials, in which metallic PCMs (gallium, Bi-based alloys) are integrated alongside organic PCMs to leverage the high thermal conductivity of metals for rapid initial heat absorption and the high latent heat of organics for sustained thermal buffering. Boteler et al. demonstrated that this approach enables thermal solutions 25–50% lighter than pure metallic systems while maintaining comparable thermal performance [19].

7. Gallium Supply Chain Constraints

Gallium is not mined directly; it is recovered almost exclusively as a by-product during the processing of aluminum from bauxite and, secondarily, from electrolytic zinc refining [26]. Its crustal abundance is approximately 16 parts per million. Approximately 95% of annual gallium consumption is directed toward gallium arsenide (GaAs) wafer fabrication for the electronics industry [26]. Gallium nitride (GaN) power electronics represent a second, rapidly growing source of competing demand.

In July 2023, China, which accounts for approximately 80% of global primary gallium production, imposed export controls requiring special licenses for shipment [27]. The controls triggered immediate price spikes and exposed the fragility of supply concentration. The EU, the United States, and Japan have since designated gallium as a critical raw material. A lifecycle analysis of global gallium flows from 2000 to 2021 quantifies the untapped reservoir: of 5,862 tonnes entering the industrial cycle, only 967 tonnes remain in active use, with 83.5% wasted and 4,446 tonnes accumulating in manufacturing and electronic waste [31].

For compute cooling specifically, the quantities required are orders of magnitude smaller than semiconductor wafer fabrication volumes. A single PlayStation 5 uses only a few grams of Galinstan. Facility-scale cooling loops are circulatory systems in which gallium recirculates indefinitely; the operative supply chain metric is make-up volume for leak and maintenance losses, not total fill volume, which substantially reduces ongoing demand relative to the semiconductor pathway where gallium is permanently embedded in wafers. The supply constraint is therefore manageable at current projected adoption rates, but it introduces a materials dependency that facility planners must monitor—particularly given the geopolitical volatility of gallium markets and the competing pull of GaN power electronics for electric vehicles, 5G infrastructure, and defense systems.

8. Third-Generation Materials: Embedded Two-Phase Cooling

8.1. The ICECool program

As AI accelerator heat fluxes approach 1 kW/cm² and localized hot spots reach several kW/cm², a fundamentally different approach is required: embedding two-phase microfluidic cooling directly within the chip substrate. DARPA's Intrachip Enhanced Cooling (ICECool) program targeted and achieved an order-of-magnitude improvement over conventional remote cooling [28].

In the ICECool program, microchannels and micropin-fin arrays were etched directly into silicon die, and dielectric coolants (HFE-7000, R1234ze, R245fa) were pumped through these structures, undergoing evaporative phase change at the heat source. Multiple research teams demonstrated chip-level heat flux removal exceeding 1 kW/cm², with hot-spot cooling approaching 5 kW/cm². IBM demonstrated this in a commercial two-socket server: microchannels etched into a production microprocessor achieved junction temperatures approximately 25°C lower than the air-cooled baseline at full power, while simultaneously reducing processor power consumption by over 10 W [28]. For heat reuse, because the coolant undergoes phase change at the chip surface, the exit fluid temperature is anchored near the refrigerant's saturation temperature. R1234ze (GWP < 1) can be operated at elevated pressures to produce exit temperatures in the 60–80°C range suitable for district heating. Microsoft collaborated with Corintis in 2025 on AI-optimized microchannels for its Azure Maia accelerator.

8.2. Where the other materials fit

CVD diamond offers thermal conductivity of 1,700–2,000 W/m·K, and Stanford demonstrated direct diamond growth on GaN transistors at 400°C in 2025. But diamond is a solid-state heat spreader, not a fluid. It manages hotspots at the transistor—the last hundred micrometers of the thermal chain—while gallium-based systems transport heat through meters of piping at the facility level. The two materials are complementary, not competitive.

Graphene-enhanced nanofluids achieve thermal conductivities of roughly 1–2 W/m·K—a meaningful 36%+ improvement over base dielectrics, but still 15–30× below gallium-based systems. For facilities that cannot justify the cost and complexity of a gallium retrofit, graphene nanofluids may represent the best near-term upgrade path within existing immersion architectures.

Hexagonal boron nitride (hBN) is the most intriguing long-term prospect because it offers electrical insulation with theoretical thermal conductivity rivaling diamond (1,700–2,000 W/m·K). Practical hBN composite TIMs currently achieve only 5–15 W/m·K—the gap between single-crystal theory and manufacturable composite behavior is substantial, and closing it is likely a decade-scale problem. The integration challenge is not choosing a single winner but designing systems that use each material where its properties are strongest: diamond at the transistor, gallium through the facility, dielectric or two-phase fluid as the immersion medium, and potentially hBN composites as the bridge between them if the manufacturing gap closes.

9. Conclusion

The environmental debate around compute has been framed as a consumption problem. This framing is incomplete. Every watt consumed by a processor exits as heat, and the materials chosen to manage that heat determine whether it is wasted or captured as a co-product with industrial value—warming homes, growing food, replacing fossil fuels, and eliminating freshwater consumption.

First-generation dielectric fluids (0.06–0.14 W/m·K) enabled the temperature gradient breakthrough: raising waste heat output from 25–35°C to 65–75°C, crossing the district heating threshold and eliminating heat pumps. Six operational deployments validate this approach with estimated CO_2 displacement of 1,400–3,200 tonnes per 1.7 MW container per year at realistic utilization rates, though independently verified emissions data exists for only one deployment. Second-generation gallium-based liquid metals (26–39 W/m·K) offer a 200–600× thermal conductivity improvement with experimentally demonstrated 60–68% temperature reduction under pulsed loads, but face engineering challenges in corrosion, electrical conductivity, indium toxicity, and geopolitically concentrated supply chains. The distinction between commercially mature TIM applications (TRL 8–9) and undemonstrated facility-scale cooling loops (TRL 3–4) must be clearly maintained. Third-generation embedded two-phase cooling, demonstrated at 1 kW/cm² under ICECool, points toward chip-level thermal management at heat flux densities that facility-level immersion cannot reach, with exit temperatures tunable for district heating integration.

Intellectual honesty requires acknowledging limits. Heat reuse does not make Bitcoin mining carbon-neutral. The renewable share is 52%, not 100%. ASIC hardware obsolescence generates an estimated 15,000–30,000 tonnes of e-waste annually. The heat-reuse benefits documented here are primarily applicable in cold climates with existing district heating infrastructure; they do not generalize to tropical or subtropical deployments.

What is clear is the trajectory. The sector that received the most environmental scrutiny—Bitcoin mining—has made the most environmental progress, pioneering renewable power sourcing, immersion cooling, and heat reuse years before the AI industry confronted the same challenges at larger scale. The materials design frontier documented here—from dielectric fluids to gallium to embedded phase-change systems—is the engineering path along which that progress will continue.

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Use of AI-Assisted Tools

The author used Anthropic Claude (Opus) for editorial revision, structural reorganization, image production, and formatting. All analytical content, materials science review, engineering analysis, and conclusions are the author's work. Source identification and selection were performed by the author; the AI tool was used to organize and cross-reference sources the author had already collected.

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Note on literature scope: The literature review was conducted across materials science, thermal engineering, energy systems, and digital asset databases. The author does not claim the reference set is exhaustive. Readers should consult Deng and Liu, Liquid Metals for Advanced Energy Applications (AIP Publishing, 2022) and Advanced Liquid Metal Cooling (Springer, 2024) for additional references.

April 29, 2026

To the Editors of Applied Energy,

I am writing to submit a Perspective Paper entitled "*Materials Design for Compute Thermal Management: From Dielectric Fluids to Gallium-Based Cooling Systems*" for consideration in Applied Energy. The article argues that cooling material selection in compute infrastructure is not merely a thermal engineering decision but an energy-systems design decision with direct consequences for CO_2 displacement, freshwater consumption, and district heating integration. The compute economy, Bitcoin mining at 138–211 TWh/yr, and AI data centers projected to exceed 1,000 TWh by 2026 can convert nearly all consumed electricity into heat. The material chosen to manage that heat determines whether it is wasted at 25–35°C or captured at 65–80°C as a co-product with industrial value. This framing positions the manuscript as an applied energy contribution rather than a component-level thermal management study.

The manuscript's central contribution is a three-generation materials framework that, to my knowledge, has not appeared in the literature. First-generation dielectric fluids (0.06–0.14 W/m·K) enabled the temperature-gradient breakthrough that made district heating integration possible without heat pumps. Second-generation gallium-based liquid metals (26–39 W/m·K) offer a 200–600× improvement in thermal conductivity with experimentally demonstrated 60–68% temperature reduction under pulsed loads. Third-generation embedded two-phase microfluidic systems, demonstrated at 1 kW/cm^2 under DARPA's ICECool program, address chip-level heat flux densities that facility-level immersion cannot reach. The article integrates this materials progression with six documented operational heat-reuse deployments and an original supply chain analysis distinguishing circulatory cooling loops, where gallium recirculates indefinitely, from consumptive semiconductor fabrication, where gallium is permanently embedded in wafers. This distinction, informed by Guan et al.'s lifecycle analysis of 5,862 tonnes of gallium flow (Sci. Total Environ., 2025), reframes the supply chain risk for facility planners.

The manuscript aligns directly with Applied Energy's stated scope in waste heat recovery, energy conversion and conservation, district heating systems, and the environmental impact of energy technologies. It bridges materials science and applied energy systems—connecting thermal conductivity at the chip interface to CO_2 displacement at the district heating grid—which is the kind of interdisciplinary contribution the journal invites. The article also addresses water consumption elimination through immersion cooling (zero freshwater vs. 110 million gallons/yr for equivalent evaporative systems) and documents the convergence between Bitcoin mining's thermal innovations and AI data center adoption of the same technologies, a pattern with direct implications for energy infrastructure planning.

My materials science perspective is grounded in primary research experience in gallium-based alloy thermodynamics. As a member of the Navrotsky group at Arizona State University, I conducted calorimetric measurements of alloys liquid below room temperature and presented this work orally at the 52nd CALPHAD conference in Busan, Korea (2025). My published work on enthalpies of mixing for Ga-based systems (J. Thermal Anal. Calorim., 2024) informs the assessment of gallium's thermal properties, phase-change behavior, and engineering challenges presented in this article. The perspective offered is that of a researcher who has handled liquid gallium in a calorimetry laboratory, not solely a reviewer of secondary literature.

This manuscript is the work of a single author. I declare no competing financial interests and hold no material positions in Bitcoin, mining companies, or energy firms discussed herein. The manuscript is not under consideration at any other journal and has not been published previously. Supplementary material containing extended analyses of grid integration, financial viability, hardware lifecycle, and the detailed CO_2 displacement model is provided for reviewer access. I welcome the opportunity to address any questions from the editorial team or reviewers.

Respectfully,

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Highlights

Materials Design for Compute Thermal Management: From Dielectric Fluids to Gallium-Based Cooling Systems

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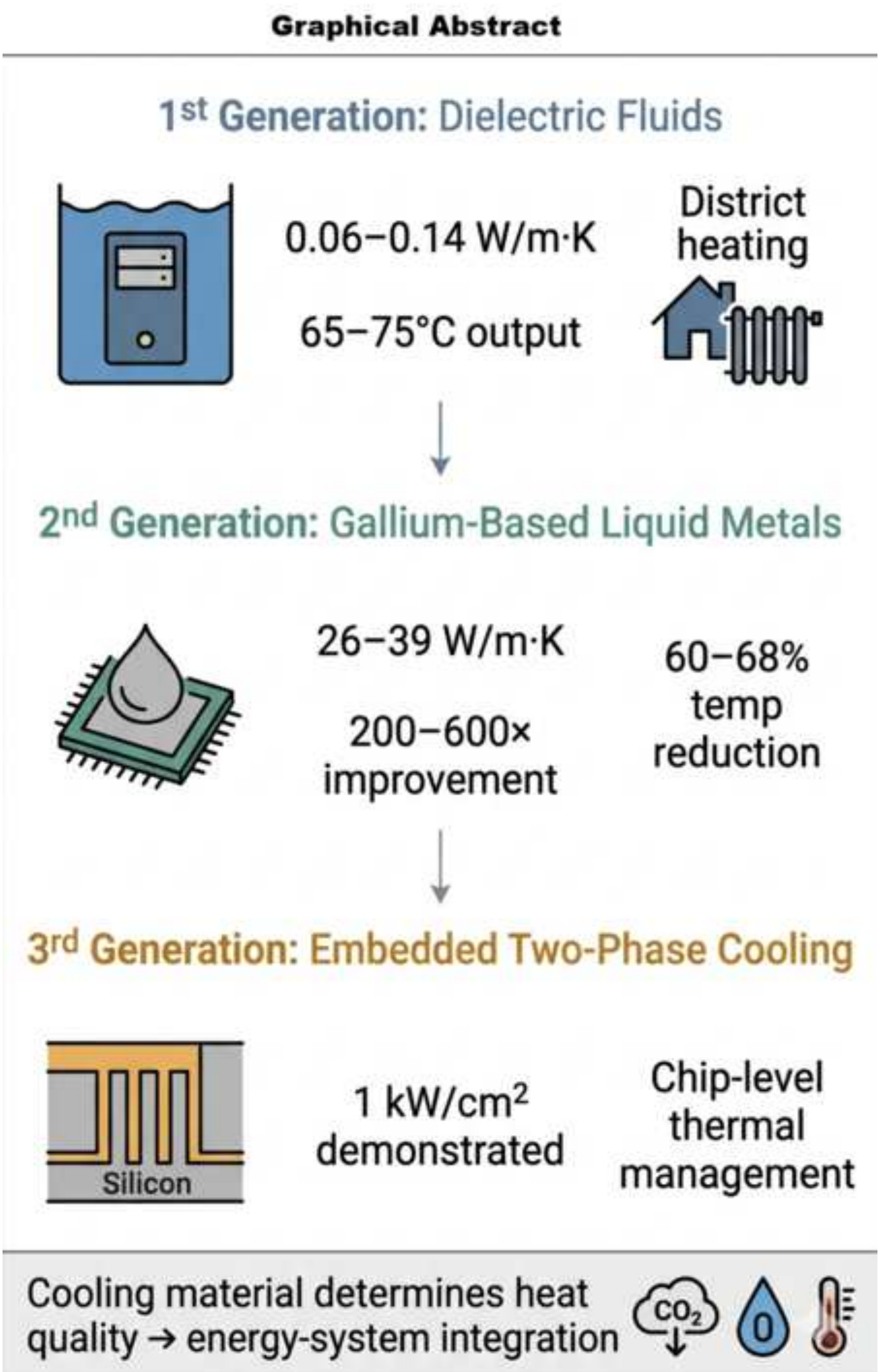
- Dielectric immersion cooling (0.06–0.14 W/m·K) captures compute waste heat at 65–75°C for district heating
- Ga-based liquid metals achieve 200–600× higher thermal conductivity with 60–68% temperature reduction
- Six operational heat-reuse deployments documented with CO_2 displacement of 2,000–3,800 t/yr
- Embedded two-phase microfluidic cooling demonstrated at 1 kW/cm² under DARPA ICECool
- Cooling material selection is an energy-systems design decision, not merely a thermal one

CRediT Author Statement

Materials Design for Compute Thermal Management: From Dielectric Fluids to Gallium-Based Cooling Systems

Michael Bustamante

Michael Bustamante: Conceptualization, Investigation, Formal analysis, Writing – original draft, Writing – review & editing, Visualization, Project administration.



Graphical Abstract
1st Generation: Dielectric Fluids
0.06–0.14 W/m·K
65–75°C output
District heating
2nd Generation: Gallium-Based Liquid Metals
26–39 W/m·K
200–600× improvement
60–68% temp reduction
3rd Generation: Embedded Two-Phase Cooling
Silicon
1 kW/cm2 demonstrated
Chip-level thermal management
Cooling material determines heat quality → energy-system integration
CO2
0

Supplementary Material

for

Materials Design for Compute Thermal Management:

From Dielectric Fluids to Gallium-Based Cooling Systems

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This supplementary material extends the main manuscript with additional analyses that were scoped out of the primary article to maintain its materials-design focus. The sections below provide the financial modeling, grid integration analysis, hardware lifecycle discussion, and extended CO_2 displacement calculations referenced in the main text. Section numbering uses the S-prefix to distinguish from the main manuscript.

S1. Grid Integration: Demand Response and Stranded Energy

This section extends the main manuscript's treatment of compute heat as a co-product by documenting the grid-flexibility dimension of Bitcoin mining, which is relevant to energy-systems planners evaluating combined heat-and-compute installations.

S1.1. Demand response

Bitcoin mining provides rapid, large-scale demand response that no other major compute workload can match. Mining operations curtail within minutes of grid signals. Riot Platforms' Rockdale, Texas facility, the largest Bitcoin mining data center in North America at 700 MW total capacity, curtailed more than 95% of its power usage during the August 2023 Texas heat wave, earning $31.7 million in demand response credits from ERCOT in a single month, more than its total credits for all of 2022. According to ERCOT, Bitcoin mining accounts for approximately 4% of forecasted peak loads [S1, S2].

Marathon Digital's CEO has stated that curtailment requests affect operations less than 3% of the time, approximately 5–10 hours per month [S1]. The economic structure is mutually beneficial: miners absorb excess renewable generation during low-demand periods, reducing curtailment of wind and solar, and release capacity during peak events, reducing blackout risk. Texas A&M University peer-reviewed research found that Bitcoin mining's load flexibility could potentially avoid all reliability concerns without major economic loss [S1]. The scale of this shift is accelerating: ERCOT large-load power requests jumped to 226 GW in 2025, nearly 4× the 63 GW recorded at end of 2024, with AI companies now accounting for 73% of new applications, displacing Bitcoin miners as the dominant flexible load requestor. Bitcoin miners pioneered the demand-response model that AI operators are now inheriting.

Qualification: Other industrial loads provide demand response, including aluminum smelting (~1–4 hour ramp-down) and hydrogen electrolysis (~30–60 minute ramp-down). Bitcoin mining's advantage is speed (minutes), operational simplicity (no process shutdown), and economic indifference to timing. It is the fastest-responding large-scale flexible load currently deployed on the ERCOT grid [S2]. Dual-revenue operation (mining + curtailment) improves project economics by an estimated 15–25% over mining-only models, making renewable-powered operations viable in locations where mining revenue alone would be marginal.

S1.2. Stranded energy and flared gas

In oil-producing regions, associated natural gas is routinely flared or vented. Portable mining containers powered by wellhead generators convert stranded gas to electricity on-site. The addressable market is estimated at $16 billion [S3].

The emissions comparison: enclosed combustion devices at wellhead installations achieve 95–99% destruction efficiency per EPA New Source Performance Standards (NSPS) Subpart OOOO/OOOOa, versus 90–98% for open flaring under favorable conditions (wind-dependent, with significant methane slip during crosswinds and low-flow events) [S4]. Where mining replaces venting (direct methane release, 80× warming potential of CO_2 over 20 years), the climate benefit is substantial. Capital cost: $500K–$1.5M per unit; payback at current BTC prices: 12–18 months.

S2. Financial Viability

This section provides the financial modeling referenced in the main manuscript's discussion of deployment scalability (Section 4.3). The heat-reuse model's economic viability is contingent on Bitcoin price, and municipal planners require sensitivity analysis across price scenarios.

S2.1. Capital and revenue structure

Container cost (Power Mining): ~€300,000 ($349,000) for 1.7 MW thermal capacity

Grid connection: $50,000–$150,000 depending on proximity

Total deployment: $400,000–$500,000 per container

Revenue streams: (1) Bitcoin mining; (2) heat revenue/avoided heating cost; (3) demand-response credits

Comparable conventional 1.7 MW boiler: $200,000–$400,000 (plus ongoing fuel cost)

Comparable 1.7 MW heat pump: $500,000–$2,000,000

S2.2. BTC price sensitivity

At $78,000 BTC: Mining revenue ~$750K–$950K/yr; payback 6–12 months; heat is supplemental revenue.

At $40,000 BTC: Revenue ~$380K–$490K/yr; payback 12–18 months; heat revenue becomes material to viability.

At \$20,000 BTC: Revenue ~\$190K–\$245K/yr; payback 24–36 months; marginal without heat or curtailment revenue.

Below \$15,000 BTC: Most operations unprofitable; system becomes an expensive boiler.

Municipalities should stress-test against the \$20,000 scenario. The model works best as supplemental heat alongside conventional or renewable systems, not as sole-source infrastructure dependent on cryptocurrency prices.

S3. Hardware Lifecycle and Alternative Consensus Mechanisms

This section addresses two challenges that reviewers of the main manuscript's heat-reuse framework are likely to raise: the electronic waste implications of ASIC hardware obsolescence, and the existence of proof-of-stake as a lower-energy alternative to proof-of-work.

S3.1. Electronic waste

ASIC miners have 3–5 year operational lifespans. An average ASIC weighs approximately 12–15 kg. With the global network operating approximately 5–7 million active devices (estimated from total hash rate of ~700 EH/s and average device output of ~100–140 TH/s), and a 3–5 year replacement cycle, approximately 1–2 million units are retired annually, producing an estimated 15,000–30,000 metric tonnes of e-waste per year. For context, global electronics e-waste totals approximately 62 million metric tonnes annually; Bitcoin mining hardware represents roughly 0.03–0.05% of this total [S5].

The heat-reuse model partially mitigates this: older hardware can remain operational as a heat source even after mining profitability declines, extending useful life by 2–3 years. However, the structural incentive of proof-of-work—continuous efficiency competition—creates an ongoing disposal stream that proof-of-stake does not.

S3.2. The proof-of-stake alternative

Ethereum's proof-of-stake transition (September 2022) reduced energy consumption by ~99.95%. The security models differ: proof-of-work secures through thermodynamic cost (physics); proof-of-stake through economic cost (capital at risk). Whether this trade-off is acceptable is a security question, not an environmental one. Bitcoin's community has chosen physics-based security, a values-driven design decision with measurable environmental consequences that the heat-reuse and demand-response models documented in this article partially offset.

S4. Extended Lifecycle CO_2 Displacement Model

This section provides the detailed calculation underlying the 1,400–3,200 tCO_2/yr displacement estimate stated in the main manuscript (Section 4.2). This is an engineering projection based on published specifications, not a measured result.

S4.1. Model parameters

Thermal output: 1.7 MW continuous

Annual operating hours (8-month heating season): ~5,840 hours

Annual thermal energy delivered: 1.7 MW × 5,840 h = 9,928 MWh

Displacement varies by fuel replaced:

If displacing natural gas: 9,928 MWh × 0.2 tCO_2/MWh = ~1,986 tCO_2/year

If displacing fuel oil: 9,928 MWh × 0.27 tCO_2/MWh = ~2,681 tCO_2/year

If displacing peat (Finland): 9,928 MWh × 0.38 tCO_2/MWh = ~3,773 tCO_2/year

The raw figures above represent theoretical maximum displacement at 100% thermal utilization. In practice, thermal utilization rates of 70–85% are typical for well-integrated district heating connections due to demand fluctuation, maintenance downtime, and network hydraulic constraints. Applying this utilization factor reduces effective thermal delivery to approximately 6,950–8,440 MWh/year, yielding adjusted displacement estimates of: natural gas: ~1,390–1,688 tCO_2/year; fuel oil: ~1,877–2,279 tCO_2/year; peat: ~2,641–3,207 tCO_2/year. The rounded range of 1,400–3,200 tCO_2/year reported in the main manuscript reflects these utilization-adjusted figures. Additionally, distribution losses of 5–15% typical for district heating networks would further reduce delivered energy; these are not included in the above estimates. The electricity consumed by the mining rigs carries its own emissions burden; at the Bitcoin network's current energy mix (52.4% sustainable), each MWh of mining electricity produces approximately 0.29 tCO_2. The net climate benefit is the difference between displaced heating emissions and mining electricity emissions.

S4.2. Infrastructure lifecycle mismatch

District heating networks are designed for 30-year lifespans. Mining hardware is obsolete in 3–5 years; data center hardware in 7–10. Municipalities evaluating mining-heated districts must plan for multiple hardware refresh cycles per infrastructure lifecycle. Heat output profiles may change with each hardware generation. This does not invalidate the model but introduces planning complexity that conventional heat sources do not [S6].

Many Finnish district heating systems already source heat from co-generation plants. Tervo et al. [S8] have shown that in these systems, the marginal climate benefit of data center waste heat is reduced because the displaced source was already efficient. The strongest case for compute heat exists where networks depend on standalone fossil boilers, not where efficient co-generation is already in place.

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